

Winfree's Conclusions

It turned out that Wang tiles on the molecular scale could indeed be constructed. Nadrian Seeman, a pioneer of DNA nanotechnology, envisioned the use of DNA as an architectural element. DNA, it turns out, can morph into structures other than the double helix. Seeman and his students built a variety of nanostructures—a wire-frame cube, a truncated octahedron, and more—all with DNA; and most especially, a four-armed 'brick' known as a double-crossover (DX) DNA molecule. This DX molecule was the molecular Wang tile Winfree was looking for. The DX molecule's four arms could be given sequences of DNA bases (A, G, C or T) corresponding to the colour labels on the four sides of the Wang tile it represented.

In a 1998 paper, *Simulations of Computing by Self-Assembly*, Winfree argued it was plausible to perform Turing-universal computation by crystallisation, or crystal growth. He summarised at the end of the paper: "Our results lend credence to proposals for computation by self-assembly of DNA: we have found that 2D self-assembly can theoretically support computation with arbitrarily low error rates."

"With arbitrarily low error rates" is science-speak for "perfectly, depending on your implementation." So, DNA can indeed self-assemble in 2D, and this self-assembly means DNA is complete computing material!

If things are a little fuzzy at this point, a little summary should make it clear:

- ☐ DNA can be nano-engineered into DX molecules, or molecular Wang tiles
- ☐ Molecular Wang tiles self-assemble into 2D crystal lattices
- ☐ Crystal lattice growth is similar to tiling the plane
- ☐ Tiling the plane means universal computing
- ☐ Hence (if Winfree's conclusions are correct) DNA can be used for universal computing!

DNA Nets

More on the self-assembly and nanotechnology front: Duke University scientists used, in 2003, self-assembling DNA molecules to build molecular meshes that could expand and contract, according to "Switchable net woven from DNA," published by *Nature*. Such a molecular mesh "might even be the basis of a computer that has as its components lumps of metals or semiconductors just a few nanometres wide. The switchable DNA net could turn electrical communication between these devices on and off by altering the distance between them." On and off—ones and zeroes! And once we build something that can manipulate ones and zeroes, we can, in theory, build a computer around it.

Today, Tomorrow, And The Day After

There are many visions for DNA computing out there. If you view it in the short term, you could say it's limited to massively parallel problems; that it is "an interesting idea, but..." If you're looking a few years ahead, you'll factor in that biotechnology is, strictly speaking, only a few years old, and that DNA manipulation techniques will take us well beyond where we are today. And if you look even further—at the work of Winfree, Seeman and others—you might see that nanotechnology is where it's at and that along with quantum computing, DNA computing is the future.

You won't see a desktop DNA computer soon, but whether you'll see one in your lifetime is an open question. If Moore's Law holds good for all of technology, you probably will. ☐

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